Section 1: Loops and Basic Programs — Q1: Print all even numbers from 1 to 20

Program

```
for number in range(1, 21):
    if number % 2 == 0:
        print(number, end=' ')
```

Output

```
2 4 6 8 10 12 14 16 18 20
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 1 — Q2: Print all odd numbers from 1 to 20

Program

```
for number in range(1, 21):
    if number % 2 != 0:
        print(number, end=' ')
```

Output

```
1 3 5 7 9 11 13 15 17 19
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 1 — Q3: Print multiples of 5 from 5 to 50

Program

```
for number in range(5, 51, 5):
    print(number, end=' ')
```

Output

```
5 10 15 20 25 30 35 40 45 50
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 1 — Q4: Find the sum of numbers from 1 to 10

Program

```
total = 0
for number in range(1, 11):
    total += number
print('Sum =', total)
```

Output

```
Sum = 55
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 1 — Q5: Find the sum of even numbers from 1 to 100

Program

```
total = 0
for number in range(2, 101, 2):
    total += number
print('Sum of even numbers =', total)
```

Output

```
Sum of even numbers = 2550
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 1 — Q6: Find the sum of odd numbers from 1 to 100

Program

```
total = 0
for number in range(1, 101, 2):
    total += number
print('Sum of odd numbers =', total)
```

Output

```
Sum of odd numbers = 2500
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 1 — Q7: Print the multiplication table of 7

Program

```
for i in range(1, 11):
    print('7 x', i, '=', 7 * i)
```

Output

```
7 x 1 = 7
7 x 2 = 14
7 x 3 = 21
7 x 4 = 28
7 x 5 = 35
7 x 6 = 42
7 x 7 = 49
7 x 8 = 56
7 x 9 = 63
7 x 10 = 70
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 1 — Q8: Take a number from the user and print its multiplication table

Program

```
number = int(input('Enter a number: '))
for i in range(1, 11):
    print(number, 'x', i, '=', number * i)
```

Output

```
Enter a number: 5
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 1 — Q9: Print numbers from 1 to 100 divisible by both 3 and 5

Program

```
for number in range(1, 101):
    if number % 3 == 0 and number % 5 == 0:
        print(number, end=' ')
```

Output

```
15 30 45 60 75 90
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 1 — Q10: Count even and odd numbers between 1 and 50

Program

```
even_count = 0
odd_count = 0
for number in range(1, 51):
    if number % 2 == 0:
        even_count += 1
    else:
        odd_count += 1
print('Even numbers:', even_count)
print('Odd numbers:', odd_count)
```

Output

```
Even numbers: 25
Odd numbers: 25
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 1 — Q11: Print list elements in reverse order

Program

```
data = [22, 33, 12, 32, 45, 13]
for item in reversed(data):
    print(item, end=' ')
```

Output

```
13 45 32 12 33 22
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 1 — Q12: Find the largest number in a list using a for loop

Program

```
data = [22, 55, 12, 89, 34, 67]
largest = data[0]
for number in data:
    if number > largest:
        largest = number
print('Largest number =', largest)
```

Output

```
Largest number = 89
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 1 — Q13: Find the smallest number in a list

Program

```
data = [22, 55, 12, 89, 34, 67]
smallest = data[0]
for number in data:
    if number < smallest:
        smallest = number
print('Smallest number =', smallest)
```

Output

```
Smallest number = 12
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 1 — Q14: Count vowels in a string

Program

```
text = input('Enter a string: ')
count = 0
for character in text.lower():
    if character in 'aeiou':
        count += 1
print('Number of vowels =', count)
```

Output

```
Enter a string: Education
Number of vowels = 5
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 1 — Q15: Count spaces in a sentence

Program

```
text = 'Python is easy to learn'
spaces = 0
for character in text:
    if character == ' ':
        spaces += 1
print('Number of spaces =', spaces)
```

Output

```
Number of spaces = 4
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 2: Conditional Statements and Nested If — Q1: ATM Withdrawal

Program

```
correct_pin = 1234
pin = int(input('Enter PIN: '))
balance = 10000
amount = int(input('Enter withdrawal amount: '))
if pin == correct_pin:
    if amount % 100 == 0:
        if balance >= amount:
            balance -= amount
            print('Withdrawal successful')
            print('Remaining balance:', balance)
        else:
            print('Insufficient Balance')
    else:
        print('Amount must be a multiple of 100')
else:
    print('Invalid PIN')
```

Output

```
Enter PIN: 1234
Enter withdrawal amount: 2000
Withdrawal successful
Remaining balance: 8000
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 2 — Q2: Student Result System

Program

```
m1 = int(input('Enter marks of subject 1: '))
m2 = int(input('Enter marks of subject 2: '))
m3 = int(input('Enter marks of subject 3: '))
if m1 >= 40 and m2 >= 40 and m3 >= 40:
    percentage = (m1 + m2 + m3) / 3
    print('Percentage:', percentage)
    if percentage >= 75:
        print('Distinction')
    elif percentage >= 60:
        print('First Division')
    elif percentage >= 50:
        print('Second Division')
    else:
        print('Pass')
else:
    print('Fail')
```

Output

```
Enter marks of subject 1: 80
Enter marks of subject 2: 70
Enter marks of subject 3: 75
Percentage: 75.0
Distinction
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 2 — Q3: Bank Loan Eligibility

Program

```
age = int(input('Enter age: '))
salary = float(input('Enter monthly salary: '))
credit_score = int(input('Enter credit score: '))
if age >= 21:
    if salary >= 25000:
        if credit_score >= 750:
            print('Loan approved')
        elif credit_score >= 650:
            print('Loan approved with higher interest')
        else:
            print('Loan rejected')
    else:
        print('Salary is not sufficient')
else:
    print('Age is not eligible')
```

Output

```
Enter age: 25
Enter monthly salary: 30000
Enter credit score: 760
Loan approved
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 2 — Q4: E-Commerce Discount

Program

```
cart_amount = float(input('Enter cart amount: '))
premium = input('Premium member? (yes/no): ').lower()
if cart_amount >= 5000:
    if premium == 'yes':
        discount = 0.20
    else:
        discount = 0.10
else:
    if premium == 'yes':
        discount = 0.05
    else:
        discount = 0
final_amount = cart_amount * (1 - discount)
print('Discount:', int(discount * 100), '%')
print('Final amount:', final_amount)
```

Output

```
Enter cart amount: 6000
Premium member? (yes/no): yes
Discount: 20 %
Final amount: 4800.0
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 2 — Q5: Login and Role-Based Access

Program

```
correct_username = 'admin'
correct_password = '12345'
username = input('Username: ')
if username == correct_username:
    password = input('Password: ')
    if password == correct_password:
        role = input('Enter role: ').lower()
        if role == 'admin':
            print('Full Access')
        elif role == 'teacher':
            print('Teacher Dashboard')
        elif role == 'student':
            print('Student Dashboard')
        else:
            print('Invalid Role')
    else:
        print('Wrong Password')
else:
    print('Invalid Username')
```

Output

```
Username: admin
Password: 12345
Enter role: teacher
Teacher Dashboard
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 2 — Q6: Movie Ticket Booking

Program

```
seats_available = True
age = int(input('Enter customer age: '))
ticket_price = 300
if seats_available:
    if age < 5:
        amount = 0
    elif age <= 12:
        amount = ticket_price * 0.50
    elif age > 60:
        amount = ticket_price * 0.70
    else:
        amount = ticket_price
    print('Ticket amount:', amount)
else:
    print('House Full')
```

Output

```
Enter customer age: 10
Ticket amount: 150.0
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 2 — Q7: Electricity Bill

Program

```
units = int(input('Enter units consumed: '))
if units <= 100:
    bill = units * 5
elif units <= 200:
    bill = units * 7
elif units <= 300:
    bill = units * 10
else:
    bill = units * 15
if bill > 3000:
    bill += bill * 0.10
print('Electricity bill: ■', bill)
```

Output

```
Enter units consumed: 250
Electricity bill: ■ 2500
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 2 — Q8: Employee Bonus

Program

```
experience = float(input('Years of experience: '))
salary = float(input('Salary: '))
performance = int(input('Performance score: '))
if experience >= 1 and salary >= 50000:
    if performance >= 90:
        rate = 0.20
    elif performance >= 75:
        rate = 0.10
    else:
        rate = 0.05
elif salary < 50000:
    if performance >= 90:
        rate = 0.15
    else:
        rate = 0.05
else:
    rate = 0
print('Bonus amount:', salary * rate)
```

Output

```
Years of experience: 3
Salary: 60000
Performance score: 92
Bonus amount: 12000.0
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 2 — Q9: Driving License

Program

```
age = int(input('Enter age: '))
learner_license = input('Learner license available? (yes/no): ').lower()
score = int(input('Driving test score: '))
if age >= 18:
    if learner_license == 'yes':
        if score >= 60:
            print('Eligible for driving license')
        else:
            print('Driving test not passed')
    else:
        print('Learner license required')
else:
    print('Age must be at least 18')
```

Output

```
Enter age: 20
Learner license available? (yes/no): yes
Driving test score: 75
Eligible for driving license
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 2 — Q10: ATM Difficult Challenge

Program

```
correct_pin = 1234
account_active = True
balance = 30000
pin = int(input('Enter PIN: '))
if pin == correct_pin:
    if account_active:
        amount = int(input('Enter amount: '))
        if amount > 0:
            if amount % 100 == 0:
                if balance - amount >= 500:
                    if amount > 20000:
                        print('Extra verification required')
                    balance -= amount
                    print('Withdrawal successful')
                    print('Remaining balance:', balance)
                else:
                    print('Minimum balance of ■500 must remain')
            else:
                print('Amount must be a multiple of ■100')
        else:
            print('Amount must be positive')
    else:
        print('Account is inactive')
else:
    print('Invalid PIN')
```

Output

```
Enter PIN: 1234
Enter amount: 5000
Withdrawal successful
Remaining balance: 25000
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 3: Dictionaries, Loops and Nested Conditions — Q1: Check Student Result

Program

```
students = {'Rahul': 75, 'Amit': 32, 'Priya': 88, 'Neha': 39, 'Rohit': 55}
for name, marks in students.items():
    if marks >= 40:
        print(name, 'Pass')
    else:
        print(name, 'Fail')
```

Output

```
Rahul Pass
Amit Fail
Priya Pass
Neha Fail
Rohit Pass
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 3 — Q2: Student Performance Using Nested If

Program

```
students = {'Rahul': 85, 'Amit': 35, 'Priya': 72, 'Neha': 48}
for name, marks in students.items():
    if marks >= 80:
        category = 'Excellent'
    elif marks >= 60:
        category = 'Good'
    elif marks >= 40:
        category = 'Average'
    else:
        category = 'Fail'
    print(name, category)
```

Output

```
Rahul Excellent
Amit Fail
Priya Good
Neha Average
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 3 — Q3: Product Stock Management

Program

```
products = {'Laptop': 10, 'Mouse': 3, 'Keyboard': 0, 'Monitor': 7, 'Printer': 2}
for product, stock in products.items():
    if stock == 0:
        status = 'Out of Stock'
    elif stock <= 5:
        status = 'Low Stock'
    else:
        status = 'Available'
    print(product, status)
```

Output

```
Laptop Available
Mouse Low Stock
Keyboard Out of Stock
Monitor Available
Printer Low Stock
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 3 — Q4: Employee Salary and Department

Program

```
employees = {'Rahul': {'salary': 60000, 'department': 'IT'}, 'Amit': {'salary': 40000, 'department': 'HR'}, 'P
for name, details in employees.items():
    salary = details['salary']
    department = details['department']
    if salary >= 50000:
        if department == 'IT':
            category = 'Senior IT Employee'
        elif department == 'HR':
            category = 'Senior HR Employee'
        else:
            category = 'Senior Employee'
    else:
        category = 'Junior Employee'
    print(name, category)
```

Output

```
Rahul Senior IT Employee
Amit Junior Employee
Priya Senior IT Employee
Neha Junior Employee
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 3 — Q5: Check Even/Odd Values

Program

```
data = {'a': 10, 'b': 15, 'c': 25, 'd': 30, 'e': 40}
for key, value in data.items():
    if value % 2 == 0:
        if value > 20:
            result = 'Large Even'
        else:
            result = 'Even'
    else:
        result = 'Odd'
    print(key, result)
```

Output

```
a Even
b Odd
c Odd
d Large Even
e Large Even
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 3 — Q6: Shopping Cart Classification

Program

```
products = {'Mobile': 15000, 'Mouse': 500, 'Laptop': 60000, 'Keyboard': 1200}
for product, price in products.items():
    if price >= 5000:
        category = 'Premium Product'
    elif price >= 1000:
        category = 'Expensive Product'
    else:
        category = 'Affordable Product'
    print(product, category)
```

Output

```
Mobile Premium Product
Mouse Affordable Product
Laptop Premium Product
Keyboard Expensive Product
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 3 — Q7: Student Eligibility

Program

```
students = {'Rahul': {'attendance': 85, 'marks': 70}, 'Amit': {'attendance': 60, 'marks': 80}, 'Priya': {'atte
for name, details in students.items():
    if details['attendance'] >= 75:
        if details['marks'] >= 40:
            result = 'Eligible'
        else:
            result = 'Failed in Exam'
    else:
        result = 'Not Eligible'
    print(name, result)
```

Output

```
Rahul Eligible
Amit Not Eligible
Priya Failed in Exam
Neha Eligible
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 3 — Q8: Bank Account Classification

Program

```
accounts = {'Rahul': 25000, 'Amit': 5000, 'Priya': 0, 'Neha': -2000}
for name, balance in accounts.items():
    if balance >= 10000:
        category = 'Premium Account'
    elif balance > 0:
        category = 'Regular Account'
    elif balance == 0:
        category = 'No Balance'
    else:
        category = 'Overdraft'
    print(name, category)
```

Output

```
Rahul Premium Account
Amit Regular Account
Priya No Balance
Neha Overdraft
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 3 — Q9: Nested Dictionary - Employee Experience

Program

```
employees = {'Rahul': {'salary': 60000, 'experience': 6}, 'Amit': {'salary': 40000, 'experience': 2}, 'Priya'
for name, details in employees.items():
    salary = details['salary']
    experience = details['experience']
    if salary >= 50000:
        if experience >= 5:
            category = 'Senior'
        else:
            category = 'Mid-Level'
    else:
        if experience >= 3:
            category = 'Experienced Junior'
        else:
            category = 'Fresher/Junior'
    print(name, category)
```

Output

```
Rahul Senior
Amit Fresher/Junior
Priya Mid-Level
Neha Experienced Junior
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 3 — Q10: Student Marks and Attendance

Program

```
students = {'Rahul': {'marks': 85, 'attendance': 90}, 'Amit': {'marks': 35, 'attendance': 80}, 'Priya': {'mark
for name, details in students.items():
    marks = details['marks']
    attendance = details['attendance']
    if attendance < 75:
        result = 'Not Eligible'
    elif marks >= 80:
        result = 'Distinction'
    elif marks >= 60:
        result = 'First Division'
    elif marks >= 40:
        result = 'Pass'
    else:
        result = 'Fail'
    print(name, result)
```

Output

```
Rahul Distinction
Amit Fail
Priya Not Eligible
Neha Pass
Rohit Not Eligible
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4: List Comprehensions — Q1: Numbers greater than 50

Program

```
numbers = [20, 55, 30, 80, 45, 90, 10]
result = [number for number in numbers if number > 50]
print(result)
```

Output

```
[55, 80, 90]
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q2: Numbers between 20 and 50

Program

```
numbers = [10, 25, 35, 60, 45, 15, 50, 70]
result = [number for number in numbers if 20 <= number <= 50]
print(result)
```

Output

```
[25, 35, 45, 50]
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q3: Numbers divisible by both 3 and 5

Program

```
numbers = [10, 15, 20, 30, 45, 50, 60, 75]
result = [number for number in numbers if number % 3 == 0 and number % 5 == 0]
print(result)
```

Output

```
[15, 30, 45, 60, 75]
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q4: Names having more than 4 characters

Program

```
names = ['Amit', 'Rahul', 'Neha', 'Rohit', 'Sneha']
result = [name for name in names if len(name) > 4]
print(result)
```

Output

```
['Rahul', 'Rohit', 'Sneha']
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q5: Names starting with A

Program

```
names = ['Amit', 'Rahul', 'Ankit', 'Neha', 'Arjun', 'Rohit']
result = [name for name in names if name.startswith('A')]
print(result)
```

Output

```
['Amit', 'Ankit', 'Arjun']
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q6: Extract vowels from a string

Program

```
text = 'programming'
result = [character for character in text if character in 'aeiouAEIOU']
print(result)
```

Output

```
['o', 'a', 'i']
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q7: Extract digits from a string

Program

```
data = 'Python123Programming456'
result = [character for character in data if character.isdigit()]
print(result)
```

Output

```
['1', '2', '3', '4', '5', '6']
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q8: Replace even numbers with Even and odd numbers with Odd

Program

```
numbers = [10, 15, 20, 25, 30]
result = ['Even' if number % 2 == 0 else 'Odd' for number in numbers]
print(result)
```

Output

```
['Even', 'Odd', 'Even', 'Odd', 'Even']
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q9: Replace negative numbers with 0

Program

```
numbers = [10, -5, 20, -8, 30, -2]
result = [number if number >= 0 else 0 for number in numbers]
print(result)
```

Output

```
[10, 0, 20, 0, 30, 0]
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q10: Apply product discounts

Program

```
prices = [500, 1200, 800, 2000, 1500]
final_prices = [price * 0.90 if price >= 1000 else price * 0.95 for price in prices]
print(final_prices)
```

Output

```
[475.0, 1080.0, 760.0, 1800.0, 1350.0]
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q11: Find employees earning more than ■50,000

Program

```
employees = [{'name': 'Amit', 'salary': 45000}, {'name': 'Rahul', 'salary': 60000}, {'name': 'Neha', 'salary'
result = [employee['name'] for employee in employees if employee['salary'] > 50000]
print(result)
```

Output

```
['Rahul', 'Neha']
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q12: Find products costing more than ■1,000

Program

```
prices = [499, 1299, 799, 2499, 999, 1599]
result = [price for price in prices if price > 1000]
print(result)
```

Output

```
[1299, 2499, 1599]
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q13: Give 20% discount to products costing more than ■2,000

Program

```
prices = [500, 1200, 2500, 3000, 1500, 5000]
result = [price * 0.80 for price in prices if price > 2000]
print(result)
```

Output

```
[2000.0, 2400.0, 4000.0]
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q14: Separate bank deposits and withdrawals

Program

```
transactions = [5000, -2000, 3000, -1000, 7000, -500]
deposits = [amount for amount in transactions if amount > 0]
withdrawals = [amount for amount in transactions if amount < 0]
print('Deposits:', deposits)
print('Withdrawals:', withdrawals)
```

Output

```
Deposits: [5000, 3000, 7000]
Withdrawals: [-2000, -1000, -500]
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q15: Students who scored 60 or more

Program

```
students = [{'name': 'Amit', 'marks': 75}, {'name': 'Rahul', 'marks': 45}, {'name': 'Neha', 'marks': 90}, {'na
result = [student['name'] for student in students if student['marks'] >= 60]
print(result)
```

Output

```
['Amit', 'Neha']
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q16: Employees from IT department

Program

```
employees = [{'name': 'Amit', 'department': 'IT'}, {'name': 'Rahul', 'department': 'HR'}, {'name': 'Neha', 'de
result = [employee['name'] for employee in employees if employee['department'] == 'IT']
print(result)
```

Output

```
['Amit', 'Neha']
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 4 — Q17: Employees earning more than ■60,000 and belonging to IT

Program

```
employees = [{'name': 'Amit', 'department': 'IT', 'salary': 55000}, {'name': 'Rahul', 'department': 'HR', 'sa
result = [employee['name'] for employee in employees if employee['salary'] > 60000 and employee['department']
print(result)
```

Output

```
['Neha']
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 5: Additional Real-World Conditional Challenges — Q1: ATM Withdrawal (Advanced)

Program

```
correct_pin = 1234
account_active = True
balance = 30000
pin = int(input('Enter PIN: '))
if pin != correct_pin:
    print('Invalid PIN')
elif not account_active:
    print('Account is inactive')
else:
    amount = int(input('Enter amount: '))
    if amount <= 0:
        print('Amount must be positive')
    elif amount % 100 != 0:
        print('Amount must be a multiple of ■100')
    elif balance - amount < 500:
        print('Minimum balance of ■500 must remain')
    elif amount > 20000:
        print('Extra verification required')
        print('Please complete verification before withdrawal.')
    else:
        balance -= amount
        print('Withdrawal successful')
        print('Remaining balance:', balance)
```

Output

```
Enter PIN: 1234
Enter amount: 5000
Withdrawal successful
Remaining balance: 25000
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 5 — Q2: Electricity Bill

Program

```
units = int(input('Enter units consumed: '))
if units < 0:
    print('Units cannot be negative')
else:
    if units <= 100:
        bill = units * 5
    elif units <= 200:
        bill = units * 7
    elif units <= 300:
        bill = units * 10
    else:
        bill = units * 15
    if bill > 3000:
        bill = bill * 1.10
    print('Electricity bill: ■', bill)
```

Output

```
Enter units consumed: 350
Electricity bill: ■ 5775.0
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 5 — Q3: Employee Bonus

Program

```
experience = float(input('Years of experience: '))
salary = float(input('Salary: '))
performance = int(input('Performance score: '))
if experience >= 1 and salary >= 50000:
    if performance >= 90:
        rate = 0.20
    elif performance >= 75:
        rate = 0.10
    else:
        rate = 0.05
elif salary < 50000:
    if performance >= 90:
        rate = 0.15
    else:
        rate = 0.05
else:
    rate = 0
print('Bonus amount:', salary * rate)
```

Output

```
Years of experience: 2
Salary: 60000
Performance score: 80
Bonus amount: 6000.0
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.

Section 5 — Q4: Driving License Eligibility

Program

```
age = int(input('Enter age: '))
learner_license = input('Learner license available? (yes/no): ').lower()
score = int(input('Driving test score: '))
if age >= 18:
    if learner_license == 'yes':
        if score >= 60:
            print('Eligible for driving license')
        else:
            print('Driving test not passed')
    else:
        print('Learner license required')
else:
    print('Age must be at least 18')
```

Output

```
Enter age: 21
Learner license available? (yes/no): yes
Driving test score: 85
Eligible for driving license
```

Note: The output shown is a sample run. For input-based programs, the entered values are examples.